

# Avishek Gupta

AI/ML Engineer

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## PROFESSIONAL SUMMARY

AI/ML Engineer with 3 years of experience building and deploying production-grade LLM systems for Fortune 500 clients including Macy's, CVS Health, MetLife, Disney, and GenD. Specialises in multi-agent orchestration (LangGraph), retrieval-augmented generation, and LLM-powered backend services, having shipped 8+ agentic platforms end-to-end from system design through production deployment on AWS (Bedrock, Lambda, CodeDeploy) and GCP (Vertex AI, GKE, BigQuery). AWS Amazon Nova Partner Demo Semifinalist (2025). AWS AI Practitioner and Azure AI Engineer certified.

## TECHNICAL SKILLS

**Languages:** Python, C, C++, SQL

**AI/ML Frameworks:** LangChain, LangGraph, OpenAI, Google Gemini, Vertex AI, Anthropic Claude, Amazon Bedrock (Nova Pro/Lite, Titan Embeddings), NVIDIA NIMs, Hugging Face

**Agentic AI & Prompt Engineering:** LangGraph multi-agent orchestration, prompt engineering & iteration, tool-use agents, custom middleware architecture, Playwright browser agents, sub-agent delegation

**MLOps & Evaluation:** Langfuse (LLM tracing & cost tracking), BigQuery-based traceability metrics, fine-tuning OpenAI models, knowledge-base grounding validation

**Databases:** PostgreSQL, MongoDB, Neo4j, Qdrant (Vector DB), Redis (caching)

**Backend & API:** FastAPI (async), Flask, Uvicorn, Pydantic, REST API design, OAuth2/JWT, CORS

**DevOps/Cloud:** Docker, AWS (Bedrock, Lambda, S3, CodeDeploy, Secrets Manager), GCP (Vertex AI, GKE, BigQuery, GCS, Workload Identity), Airflow, CI/CD

**Integrations:** Jira (REST + Forge Apps + Xray), Confluence, Collibra, Tableau, Playwright, Tree-sitter

## EDUCATION

**National Institute of Technology Durgapur**

*Master of Technology (MTech) in Computer Science*

Durgapur, India

2021 – 2023

**University College of Engineering and Technology, Vinoba Bhave University**

*Bachelor of Technology (BTech) in Computer Science and Engineering*

Hazaribagh, Jharkhand

2017 – 2021

## PROFESSIONAL EXPERIENCE

**GenAI Backend Developer**

*Factspan Analytics*

Sep 2023 – Present

*Bengaluru, Karnataka, IN*

- Built and shipped 8+ production agentic AI platforms end-to-end for Fortune 500 clients (Macy's, CVS Health, MetLife, Disney, GenD) across retail, insurance, and healthcare, deployed on both AWS and GCP; led the Flux platform collaboratively in a 3-person team.
- Designed a reusable middleware-based agent architecture, adopted across 4+ projects, that exposes typed Python tool functions to LLM agents, enabling rapid assembly of new multi-agent systems from shared capabilities (file I/O, code analysis, Playwright browser automation, sub-agent delegation) instead of per-project prompt scaffolding.
- Enforced production-grade reliability across all platforms: OAuth2/JWT RBAC scoped by department, role, and region, cost-aware model-routing tiers, Langfuse observability for per-request cost and latency tracking, and isolated multi-tenant infrastructure with zero cross-client data leakage.

## KEY PROJECTS

**CatalogGuard AI** | AWS Nova Partner Demo Semifinalist | GenAI Engineer / Solution Architect

- Designed a multi-node LangGraph audit pipeline with parallel state graph execution, auditing **10,000+ products in under 10 minutes** across 4 dimensions: style compliance, visual analysis, accuracy scoring, and duplicate detection.
- Built a multimodal analysis engine using Amazon Nova Pro (Bedrock Converse API) for product image auditing, plus a configurable style compliance checker driven by 292-line YAML rule definitions covering banned superlatives, non-inclusive language, tone, and category-specific rules.
- Deployed as a serverless async FastAPI application on AWS Lambda via Mangum, containerized with Docker, with CORS-enabled REST endpoints for file upload, audit triggering, and result retrieval.

**Flux / Insights Co-Pilot Platform** | GenAI/ML Engineer (Lead) | CVS Health, Macy's, Disney, GenD

- Architected a Text-to-SQL agent translating natural language into SQL against enterprise PostgreSQL databases using a hybrid semantic layer with hot-swappable YAML and Colibra backends, enabling **42 business measures across 9 tables** and **200–300 queries/day** in production.
- Built an Analyst Co-Pilot (text-to-SQL + follow-up Q&A + data visualization + automated insights), a Compliance Analysis engine cross-referencing policy manuals against federal CFR regulations, and Document Q&A with domain-specific variants for HR, PBM, and Contract documents using Qdrant retrieval and GPT-4o.
- Deployed on AWS using Docker, CodeDeploy, TLS/SSL, Redis caching, and Uvicorn behind a reverse proxy; enforced per-user query quotas and RBAC-scoped data access across all client tenants.

#### **AI Product-Delivery Platform (Atlassian) | GenAI Engineer / Full-Stack | Macy's**

- Built an event-driven multi-agent platform closing the loop from requirements to delivery: a Confluence BRD is analysed by LLM agents into Jira epics, stories, and test cases, with every artefact recorded in a versioned knowledge graph and surfaced back as contextual overlap recommendations inside Jira/Confluence.
- Shipped two frontends over one shared FastAPI/LangGraph backend: a React 18 + Vite SPA with streaming agent chat and a multi-surface Atlassian Forge app (UI Kit 2 + Custom UI) spanning Jira panels, Confluence macros, and global pages, with per-endpoint live/mock switching for zero-redeploy parallel development.
- Engineered a multi-pass knowledge-graph pipeline with cost-aware model-routing tiers (cheap/mid/deep) cutting cost **~10x** vs single-model, plus resumable per-concept writes; served via nginx-gated FastAPI microservices behind a single HTTPS host to satisfy Forge egress rules.

#### **AI DevOps Migration & Code Estate Platform | GenAI Engineer / Backend Developer | CVS Health**

- Built a hierarchical multi-agent system (LangGraph, ReAct pattern) automating end-to-end repository migration from Bitbucket/GitLab to GitHub across a **13-step orchestrated workflow**: mirror clone with full history, CI/CD conversion to GitHub Actions, branch-protection rulesets, and post-migration verification with human-approval gates.
- Designed a composable middleware tool-injection framework (**~35 modules**) with an LLM self-healing loop for generated CI YAML; delivered code-estate intelligence via Tree-sitter AST analysis across 10+ languages, Gitleaks secret pre-scan, and IaC/K8s infrastructure mapping for migration-readiness scoring.
- Engineered a concurrent bulk-migration engine (CSV/Excel intake, thread-pool workers, SQLite state machine, pause/resume/stop, crash-resumable checkpoints) with tiered model routing, Qdrant code-Q&A retrieval, and Langfuse per-run cost accounting.

#### **Agentic Test Generation & QA Platform | GenAI Engineer / Full-Stack | Macy's**

- Reduced test-case creation cycle from **2–3 hours to under 3 minutes** by building a multi-agent LLM platform (LangGraph, Vertex AI) with **15 specialized agents** across a 5-stage pipeline that auto-generates executable pytest suites and business/SIT test packs grounded in Jira, Confluence, and design documents.
- Designed a custom middleware toolkit (file/search/ripgrep/code-skeletoniser, sub-agent spawning, Playwright browser automation, Jira/Confluence connectors) plus a grounded retrieval layer with embeddings, selective OCR, and spreadsheet/diagram parsing to keep generated tests traceable to source requirements.
- Shipped as both a standalone React 18 + Vite web platform (SSE streaming, BigQuery persistence, GCS with signed URLs, deployed on **GKE Autopilot** with Kustomize and keyless Workload Identity) and as an Atlassian Forge app with Xray test-management sync for one-click Jira story enrichment.

## CERTIFICATIONS

**AWS Certified AI Practitioner | AIF-C01**  
**Microsoft Certified: Azure AI Engineer Associate**

Sep 2025  
 Mar 2026

## AWARDS & RECOGNITION

**AWS Amazon Nova Partner Demo Competition | Semifinalist**

2025

- Built CatalogGuard AI, an AI-powered e-commerce catalog auditing platform using Amazon Bedrock. Selected as semifinalist from a global pool of AWS partner submissions.